How do Terrorist Cells Self-Assemble: Insights from an Agent-Based Model of Radicalization[1]

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Since the 9/11 attacks, there has been a steady increase in the incidence of self-starter terrorist plots in both Western Europe and the United States. Past studies of self-starter terrorism focused on how persons come to acquire radical beliefs but take the next step for granted - the transformation of isolated radical individuals into cells or networks that produce violent collective action. To better understand this second step we operationalized radicalization as precise network measures and distinguished between several kinds of radicalization scenarios. We then constructed an agent-based computational model of radicalization and validated its predictions with empirical data. Based on this model we characterized the factors that contribute to radicalization as well as deradicalization. Finally, we extended radicalization theory by identifying specific network mechanisms such as occlusion, encapsulation, and magnet-rewiring by which radicals self-organize. Our findings may allow us to distinguish the conditions that lead to small transient cells from conditions that lead to large long-lasting terrorist organizations.

Keywords: *radicalization; political violence; terrorism; self-starter terrorism; agent-based model; social networks*

[1] The authors are listed alphabetically to indicate equal co-authorship. We would like to thank Michael Macy, David Strang, and Sarah Soule for helpful comments.

A long-standing question of political violence research has been the process of radicalization - how do individuals and groups develop and act on the belief that violence is a legitimate instrument to achieve political goals (See Crenshaw 1988; Della Porta 1995; Hudson and Majeska 1999; Post, Ruby, and Shaw 2002)? There is considerable work on how radical beliefs are acquired on the individual level(Bandura 1998; Holloway and Norwood 1997; Moghaddam 2005)[2]. However, much less is understood about how individuals with *already developed* radical beliefs find others with similar views and engage in violent collective action. Indeed this process has been largely taken for granted even by scholars who acknowledge the importance of the group for radicalization.

We begin by discussing the problem of home-grown self-starter terrorism and the "search and assembly" problem of radicalization. In section two we offer a novel way to both conceptualize and measure radicalization using social network analysis. Section three explains the agent-based model of radicalization used to investigate the process, the choice of parameters, and the principles underlying the model. In section four we present the analysis and findings of the model along with an empirical validation of some of the model's predictions. We also discuss key mechanisms generating the empirical patterns such as occlusion, encapsulation, and magnetic-rewiring. Section five offers a more general discussion of the results and section six suggests future directions in this research program.

The Challenge of Home-Grown Self-Starter Terrorism

Scholarship on radicalization experienced a shift in focus from the study of more

[2] We want to emphasize that by "radical" we mean "extremist" beliefs in support of the use of violence for political goals. We do *not* mean by "radical" as having any concrete political beliefs (*e.g.* radical vs. reactionary). Hence a radical may be a person who believes in maintaining the status quo by violent means just as much as a person who believes in overthrowing the status quo by violent means.

traditional recruiter-centered radicalization towards the study of self-radicalization. The shift is driven by two recent trends in terrorism: 1) the re-emergence of the so-called "home-grown terrorism" – terrorist acts that are carried out by groups whose membership is composed entirely or predominantly of the native-born citizens of the country that is being attacked and 2) "self-starter terrorism" – terrorist acts that are carried out by small groups of individuals that don't seem to be recruited, directed, trained, or financed by any existing terrorist organization or state security agency and form more or less spontaneously through the initiative of their members. These trends have been noted by both scholars (Kirby 2007; Vidino 2007; Kohlmann 2008; Krueger 2008) as well as security agencies and police organizations[3].

Since 9/11, homegrown self-starter terrorism (HST) has become especially pronounced within Islamist terrorism – that is terrorism motivated by Muslim fundamentalism. Even Al-Qaeda has shifted to a more decentralized form (Sageman 2004; Sageman 2008), and this trend might accelerate following the recent death of its founder. This decentralization has also been noted in affiliate organizations such as Jemaah Islamiyah (Magouirk, Atran, and Sageman 2008). A rise in Islamic home-grown self-starter terrorism is probably driven by the increasing pressure on Al-Qaeda and the poor integration of Muslims in majority non-Muslim states. As a testament to this trend, a number of plots in the last five years (in the West alone) were cases of home-grown, self-starter terrorism: Netherlands (Hofstad Group 2003), Britain (London Attacks 2005; Tubes/subway plot 2005, Glasgow Airport plot 2007), Germany (2006 World Cup plot, 2007 US Airbase plot 2007), Sweden (US embassy plot 2006), Canada (2006 truck bomb plot). The United States is no exception. A RAND report identified 46 cases of "self-recruitment" to terrorism between 2001 and 2009, with 13 occurring in 2009 alone (Jenkins 2010, vii). Most of

[3] For example see (House of Commons 2006; Silber and Bhatt 2007; Hulst 2006).

these plots were intercepted but some cells have succeeded in carrying out an attack and inflicting large casualties. The most notable is the July 7th, 2005 London Bombings in Great Britain which killed 56 people (including the bombers), and injured more than 700 (House of Commons 2006)[4] and the 2004 Madrid bombings, that killed 191 and injured more than 2000 people[5]. Other "successful" attacks carried out by HST cells outside the West, since 9/11, were the 2003 Casablanca bombings in Morocco that killed 45 and injured more than a 100, the 2004 Taba hotel bombings in Egypt that killed 34 and injured 170, and some attacks in Pakistan and Iraq (see (Benjamin and Simon 2005)).

Islamist terrorism may be transforming (or expanding) into a model of terrorism long-used by right-wing terrorism and environmental terrorism, that of *leaderless resistance*. For example, the most infamous right-wing act of terrorism, the 1995 Oklahoma City Bombing was an attack carried out by just two people; but they were propelled by a large movement that created, distributed, and legitimated a very specific ideology, through pamphlets and other literature (more recently through the Internet). Extreme right wing movements such as the American Nazi Party and Christian Identity came to the conclusion that they would not be able to wage an organized military campaign but would indirectly encourage sympathizers to initiate "small acts of violent rebellion" (Kaplan 1997). A similar strategy has been used by members of the Earth Liberation Front (ELF) which actively encouraged the formation of new and independent cells. ELF's publicist, Craig Rosebraugh, summarizes the idea of leaderless resistance in an introductory video, saying "There's no realistic chance of becoming active in an

[4] There is some debate regarding the connection of the Madrid and the London cells to Al-Qaeda. A House of Commons investigation (House of Commons Report 2006:19-20) concluded that there is no evidence for the claim that the July 7th bombing has been directed by Al-Qaeda.

[5] Despite early speculations, a two-year investigation by the Spanish government concluded that there is no evidence for an Al-Qaeda connection(Haven 2006). Additionally not all the collaborators in the Madrid attack were home-grown radicals as some involved were foreign nationals. It is also not clear to what extent a formal organization was behind the attacks.

already existing cell… Take initiative; form your own cell." (Joosse 2007). It is important to note that in all of those cases there was a large community of sympathizers that never engaged in direct violence but who contributed to the cultural repertoires necessary to sustain the movement. A similar community of "cultural entrepreneurs" sustains the Islamist movement by producing books that provide highly militant interpretations of Islam, graphic videos of violent victimization of Muslims by non-Muslims[6], and websites that provide a community to interpret and reframe current events.

HST poses unique challenges to counter-terrorism policymakers. Terrorist attacks carried out by foreign nationals allow for easier detection, deeper monitoring, and greater surveillance than attacks carried out by citizens whose rights to privacy, association, and travel are constitutionally protected in most democracies. Terrorist attacks carried out by known organizations can be monitored and tracked, and ultimately stopped by incapacitating most members of those organizations. However, cells started by a small group of unknown individuals are a lot more difficult to anticipate, monitor, and counteract. Thus, incapacitating one cell does not significantly contribute to destroying another fully independent cell.

Of course not all HST cells have this short-lived and independent nature. Over time some do develop into large and sophisticated terrorist organizations. For example, the Egyptian and Saudi groups that coalesced into Al-Qaeda formed in roughly this way (Wright 2006). The two types of radicalization trajectories might be distinguishable using our model, as we explain later.

The Search and Radicalization Processes of HST

The assembly of HST terror cells involves two interacting processes: the search process

[6] In a remarkably similar form of dramatizing in-group victimization by the out-group, right-wing extremists vividly describe interracial violence of blacks against whites (Kaplan 1997).

where existing radicals find one another, and the radicalization process in which they develop increasingly radical views. These are not unique to political violence: in Microeconomics, analogous processes of search and training are inherent in the labor market, where they correspond to frictional and structural components of total unemployment.

Case studies of self-starter terrorism reveal that the individuals who join groups have some pre-existing beliefs or at least sympathies to the ideology of the group they join[7]. There is little doubt that additional radicalization happens at the group level as well. But clearly, for the group to exist in the first place, the radical sympathizers must first find one another. We are not arguing that the search is performed with equal zeal by all the radicals. Indeed more often the case is that some members are more entrepreneurial than others and play a recruiter function as was the case with Mohammad Sidique Khan. Nonetheless we maintain that all home-grown self-starter cells at some point and to some degree face a search problem. Understanding the factors that influence how they solve this problem is essential if one is to comprehend the full picture of radicalization.

In HST, search and radicalization are not necessarily sequential in the sense that radicals acquire extreme beliefs independently and then meet like-minded colleagues and then engage in violence. Instead individuals who are only sympathetic may be influenced to more extreme opinions by their friends after they join (Sageman 2004). Members in small groups have a tendency to reinforce each other's views and even further radicalize one another by outbidding each other's extreme opinions (See a review of this phenomenon byIsenberg 1986).

Conceptualizing and measuring this radicalization is the question we turn to next.

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[7] This was clearly the case with the 7/7 Bombers (House of Commons 2006, 15-18); the Hofstad Group ((Vidino 2007, 581), and some of the Islamist networks in Spain ((Jordan and Horsburgh 2005). Indeed we could find no case of self-starter home-grown terrorism where "brainwashing "or coercion of previously unsympathetic individuals has taken place.

CONCEPTUALIZING AND MEASURING RADICALIZATION

A critical objective for our study, and one that has been largely unexplored in previous work, is how to measure radicalization in the context of social networks. The most basic way to measure radicalization is to simply count the fraction of radicals within the relevant community. However this approach is problematic because it does not take the relationships between the radicals into account. There may be an increase in the total percentage of radicals but not in the number of connected cells that these radicals form (and vice versa). Because the network configuration plays such a decisive role we developed three measures or metrics of radicalization in the framework of social network analysis. Those measures are based on established ideas of social network theory. Combined together they paint a detailed picture of whether radicalization is occurring and how. We then identify distinct radicalization scenarios that are defined based on the different combinations of the three basic measures.

Network Measures of Radicalization: Isolation, Clustering and Mean Cell Size

Our first measure, termed *Isolation*, describes the connectivity between radicals and non-radicals. Low isolation represents a condition where radicals are well-connected to the society at large including many non-radicals (Figure 1, boxes A and B). High isolation represents a condition where the radicals constitute an isolated subnetwork with no ties to non-radicals (Figure 1, boxes C and D). High isolation has the greater threat of violence since the radicals experience no restraining influence from non-radicals. A suitable measure of isolation is the difference between the number of radical-to-radical ties (“internal ties”) and radical-to- non-radicals ties (“external ties”) divided by the total number of ties. Namely we measure isolation

using the E-I index (Krackhardt and Stern 1988) multiplied by -1.

Figure 1 about here

However, the full picture of radicalization requires additional measures because it usually takes more than a single radical-to-radical tie, i.e. more than two individuals, to form an effective cell. Part of this picture is painted by *Clustering* which measures the degree to which the radical community is organized into triads, in essence whether the community is "closely knit" into cliques that could be the precursors of active terrorist cells. The distinction between low and high clustering can be further seen in Figure 1; boxes A and C represent scenarios where there is very little clustering between the radicals, while boxes B and D represent a scenario where the radicals are highly clustered. To measure *Clustering* we use the $\tilde{C}$ metric from Soffer (2005) applied to the subnetwork containing just the radicals.

Finally, it is important to consider the scale of the radical community, namely, the sizes of the cells, a metric we call *mean cell size* or just *cell size*. In Figure 1 box E illustrates a network with a low mean cell size; while box F illustrates a high mean cell size (while isolation and clustering are the same).

We have also tried alternative measures of clustering and isolation which were highly correlated to the metrics we selected and did not change the analysis. The three metrics we selected show a very low correlation among themselves (see Table 1). These metrics provide information about the state of society at a particular point in time. In a dynamic society or a social simulation the values of the measures change over time, and we captured this by averaging them over the course of the simulation.

Table 1 about here

Types of Radicalization and De-radicalization

Through the metrics one can define four kinds of scenarios: 2 kinds of radicalization and 2 kinds of de-radicalization, summarized in Table 2.

Table 2 about here

The *Lone Wolfs* radicalization scenario is populated by radical individuals organized into small cells that are highly isolated from the rest of society. They are not embedded in groups of like minded individuals, and indeed they have very few friends of any kind. They may correspond to individuals such as Ted Kaczynski, Timothy McVeigh, or Eric Rudolph. Operationally a world of such individuals, in terms of our metrics of radicals, is characterized by high isolation, low clustering, and a low cell size. In this scenario, the bulk of the terrorism risk comes from Lone Wolfs rather than larger formations, even if many or most of the radicals would never "bite", i.e. pick up arms and commit violence.

The *Wolf Packs* radicalization scenario is populated by radicals who form large, internally-clustered cells while maintaining high social isolation. Wolf packs enjoy the typical benefits of collective action, perhaps producing some cells that are able to execute relatively sophisticated, high-casualty attacks that are unfeasible if each radical operated alone. They

correspond to cells such as the 7/7 bombers as well as the Madrid and Hofstad groups. Operationally a world of such individuals, in terms of our metrics of radicals, is characterized by high isolation, high clustering, and a large cell size.

The *Trapped Wolfs* scenario is populated by lone radicals who are embedded in a network of non-radicals. They may be individuals who hold radical beliefs and perhaps even considered violent action, but who have a dense network of friends indifferent to radical ideology or actively oppose it. Perhaps these are relatives that impose familial obligations on the radicals. The typical Trapped Wolf radical may wish to act out his beliefs but it would mean leaving his family destitute and bringing shame on his kin. Once again the claim is probabilistic: a Trapped Wolf may still "escape" but this escape will be less likely under conditions of being embedded. Operationally a world of such individuals, in terms of our metrics of radicals, is characterized by low isolation, low clustering, and a low cell size.

The *Trapped Wolf Packs* scenario is populated by radical groups who are embedded in a network of non-radicals. This is likely the most common empirical scenario for Islamist groups in Western countries today. As cited earlier, we know that there are very large numbers of persons who hold highly extreme beliefs; some of whom coalesce into groups. What prevents them from acting? A likely explanation is a dense network of ties and obligations to other members of their community who are not radicals: either co-workers, friends, family members, significant others, neighbors, and community leaders. Operationally a world of such individuals, in terms of our metrics of radicals, is characterized by low isolation, high clustering, and a large cell size.

METHODOLOGY

The Agent-based Modeling Approach

Terrorist cell formation can be investigated using a variety of methodologies, each having a set of strengths and weaknesses. Traditionally, this question has been addressed through investigation of the histories of defunct terrorist groups, interviews of captured members regarding how they met one another, and examination of autobiographies of terrorists reflecting on the early days of their organizations. These methodologies can provide a wealth of important information, but they also have limitations. In general, there is a shortage of empirical sources, as well as difficulties in generalizing beyond the cases examined.

We employ yet another research strategy – one of using empirically-based computer simulations. Simulation methods make it possible to describe interconnected microscopic processes in great detail and then to observe their macroscopic effects. This makes simulations well-suited for studying psychological and social processes, because these are often related in non-linear ways and cause long-run cumulative effects (Macy and Willer 2002). Therefore, simulation methods should be very fruitful in studying how radical groups self-organize because a large portion of this behavior appears to be driven by local interaction of agents, who are also affected in complex by with ecological variables. Furthermore, by creating artificial societies one can systematically manipulate the parameters of interest to discern meaningful relationships and isolate factors that will be influential over the long-term.

The use of agent-based models to study radicalization is not new: similar works include the Civil Violence Models (Epstein 2002), Threat Anticipation Model (MacKerrow 2003), NetBreaker (North, Macal, and Vos 2004), Seldon (Berry et al. 2004) and others. Our model is different in several respects, most importantly in its use of the Hopfield social network (Hopfield 1982; Hopfield 1984; Macy et al. 2003; Nowak et al. 1998) and its focus on the formation of

cells primarily around *existing* radicals. The model is also one of the only fully disclosed and open-source code models in the radicalization literature.

Model Parameters

How do individuals with radical beliefs find one another and form a group which carries out terrorist attacks, in the absence of a pre-existing organization or a coterie of recruiters? To answer this question, one has to recognize that the way individuals with radical beliefs find one another and form groups is no different from the way any persons with similar beliefs or interests find each other and form groups, be it reading groups, transient social movements, bible study societies, or swingers clubs. And since group formation has been studied quite extensively, it is reasonable to use findings from this literature as the scaffolding of our model.

Social groups can form around a great variety of interests or activities, but the process of group formation is shaped by three kinds of constraints: structural, individual, and network-related. We will first identify and discuss these categories and then operationalize them in the simulation as processes and parameters.

The structural constraints stem from the fact that new associations form within an existing social milieu (Feld 1981; Feld 1982). These new ties are shaped by factors such as geography, neighborhood, family social circles, organizations, and institutions one is in regular contact with (i.e. one's school, place of work, or religious establishment). A person who grows up in a wealthy suburb will be exposed to an entirely different social structure than a person who grows up in a poor inner-city neighborhood.

However, within the social networks that are available to any given individual, there is wide latitude as to who s/he chooses to associate with. Here the factors that influence association are internal to the individual. The network that one chooses to join is influenced by one's tastes,

values, beliefs, and resources. People seek to form durable social connections to those who are similar to them, a general process called "homophily" (see McPherson, Smith-Lovin, and Cook 2001 for a review).

Yet there is a final constraint having to do with the network itself. Because of time, cognitive resources, and social constraints each person can only have so many friends; they have a "degree budget" – the maximal amount of friends one can have. Although individual attributes are responsible for the formation of networks, the network exerts influence back on the individual. For example, those who formed a friendship on the basis of liking a particular sports team may develop a common belief on politics. Researchers have long observed that the preferences and political views of individuals are influenced by their friends and enemies.

A major challenge has been to determine the values chosen for the parameters in our model, some of which cannot be known very precisely or vary widely across cases. To arrive at general conclusions about self-radicalization, we defined a plausible parameter space, assigned distributions to the different values and conducted simulations for societies at various randomly-chosen points of the space (This process will be further discussed in the findings section of particular parameters.) The simulation could also be used to examine particular cases of radicalization by matching its parameters to that particular society (Newcomb 1957; 1943; Friedkin 1998).

Structural Parameters

A sizable literature has called attention to the relationship between the size of human aggregates and various sociological processes (Mayhew and Levinger 1976). Therefore a basic structural factor to take into account is population size which we modeled as a single parameter

denoting how many agents are interacting. We allowed for a distribution of beliefs in political extremism, beliefs which we classified into three categories: radicals, moderates, or pacifists. The parameters *fraction initially radicals* and *fraction initially pacifist* (and implicitly, the fraction initially moderate) controlled a person's initial belief but that belief was subject to change during the course of the simulation. These parameters can be calibrated based on polling data. For example according to one poll of Islamist in the UK, "15% of British Muslims agreed that violence against civilians, to defend Islam is sometimes justified" (Pew Research Center 2006, 3).

Next, we incorporated the presence of sites and various social organizations where people meet and form friendships. This is one of the most important factors in the model. Indeed, a House of Commons report on the 7/7 Bombers states that "A common factor for all 3 – Khan, Tanweer and Hussain – was the social life around the mosques, youth clubs, gyms and the Islamic bookshop in Beeston." (House of Commons 2006, 15). The report further states that "camping, canoeing, white-water rafting, paintballing and other outward bound type activities are of particular interest because they appear common factors for the July 7th bombers and other cells disrupted previously and since" (ibid, 17). Mohammed Bouyeri, Jason Walters and Samir Azzouz of the Hofstad group met at a phone shop "that was used by a group of young Muslims to pray, watch gory tapes of terrorist actions, and talk about their desire to participate in jihad" (Vidino 2007, 582). It is not surprising that would-be radicals would meet one another through preexisting groups and organizations. This view is also supported by the research mobilization literature (Jenkins 1983; McCarthy and Zald 1977). In our terminology, all sites that attract individuals and rewire their social connections by introducing them to new friendships are

referred to as *magnets*[9].

Upon examining case studies of Islamist self-starter cells, there appears to be two types of social sites of where the cell members met: 1) *radical magnets* that attract only individuals with extreme views such as mosques led by radical clerics, informal radical seminaries, bookshops and video stores specializing in fundamentalist material; 2) *neutral magnets*, which are locations that attract individuals without distinction of political views such as athletic clubs, pubs, or social centers. We also included pacifist magnets – a social site that only attracts pacifists. Indeed, there are a number of organizations that promote coexistence[10] but are unlikely to attract radicals or uncommitted moderates. We included them in order to study the effectiveness of pacifist sites as counter-radicalizing factors.

In modeling magnets we had to specify two additional properties: *magnetic exposure rate* and the *magnetic encounter rate.* The former indicates the fraction of the population that actually attends any given magnet. The latter is the probability per visit with which an individual would be introduced to another attendee[11]. In practical terms, the magnetic exposure parameter may correspond to the unemployment rate: under conditions of high unemployment, magnets of all types are likely to see greater attendance because individuals would tend to have more time to pursue their interests and more time for leisure.

Finally, an important factor to consider in a model of tie formation is the number of

[9] Note that magnets bear some resemblance to foci(Feld 1981). Foci are elements of the social structure that channel association along specific traits resulting in homophily (workplace, neighborhood, city). However, not all foci are magnets. The key property of a magnet is that it has the potential of *re*-wiring ties. Certain foci may "bring homogenous sets of people together" (Feld 1982, 798) but not really change the ties (e.g. neighborhoods, families), instead stabilizing them and actually keeping them from being rewired. Second, foci seem to require a pre-existing attribute that organizes people but such an attribute is not required of magnets. For example, neutral magnets may organize interaction by pure chance (e.g. change meeting at a neighborhood grocery store or a festival). In our conception, a person could visit a neutral magnet based on a kind of focal point strategic reasoning (a la Shelling) or purely out of accidental causes e.g. spontaneous desire for entertainment.

[10] For example the Ansar Youth Project, see http://www.ansaryouth.org.uk/main/

[11] A possible extension of the model would be to allow realistic variation between magnets along those parameters

individuals who are leaving and entering the community, the *attrition rate*. We suspected that outmigration may have a little-noted but important disruptive effect on the formation of radical cells. An additional parameter in our model, *radical attrition increment (RAI)*, allowed us to specifically vary the attrition rate of radicals. We believe that their migration rate may be greater than the population as a whole because radical individuals may find themselves maladapted to prosper in their community, and thus may have an increased tendency to migrate away. Another plausible reason to differentiate the attrition rate for radicals from the overall population is that radicals may get arrested if they act on their beliefs and get caught. The effect of *RAI* was found to be qualitatively similar to the effect of the overall attrition rate.

Individual Parameters

Individuals in our model had two kinds of attributes. First, there are fixed characteristics about a person that cannot be changed easily such as race, gender, nationality, personality, and religion. Second, human beings also have attributes that are variable such as interests, hobbies, and more mundane preferences. The overall variability of attributes in the population is determined by a parameter termed *diversity*. A particular kind of variable attribute is political beliefs, which we call *zeal*. Zeal ranges from pacifist to radical, i.e. supporting political violence.

While in practice individuals may have hundreds of these characteristics, only some of them are likely to be socially salient to the formation of ties in a given community. There might be some variability even within the salient parameters. Therefore, the parameters *fixed attribute salience* and *zeal salience* control the weight of fixed attributes and of zeal relative to variable attributes.

Another individual parameter of the model is the rate at which individuals conform or

adopt the opinions of their neighbors on average, a parameter we called the *pressurability*. Of course, this is only possible in variable attributes.

Network Parameters

As mentioned previously each person has a degree budget, the maximal amount of friends they can have. Marsden (Marsden 1987) identified the social networks of Americans from the General Social Survey at a maximum of 5, the average of kin members at 1.5, and that of non-kin at 1.4 persons[12]. New ties are formed during the course of the simulation, but once the total number of ties exceeded an individual's degree budget, the weakest one is dropped. Another important network process is triadic closure – the formation of ties between individuals that share friends (see Davis 1970). On the societal level one can think of this as the propensity to form close-knit communities even among strangers. The likelihood of this process may vary from society to society and so we controlled this process by the parameter of *transitivity*. Finally, the strength of newly-formed ties reflects the trust of strangers in a society, and is controlled by the parameter *average initial tie strength*.

Technical Description

In this section we provide additional technical information about the model. In our model, we represent a community of N individuals each possessing a set of personal attributes, rules of interaction, and ties to other individuals in the community. All agents are assumed to have the same types of attributes (for simplicity). Of those attributes, some are fixed and some are variable. For simplicity, we assume that attributes can be in one of two states, which we represent as +1 or -1. Also, each individual has a political opinion on the question of political

[12] McPherson et al (McPherson, Smith-Lovin, and Brashears 2006)finds somewhat lower numbers, while studies relying on online social networks or other digital information sources tend to show larger numbers.

violence (“zeal”): either strongly supportive (“radical”) or strongly opposing (“pacifist”) or a centrist position (“moderate”). These states are represented as +1, -1 and 0, respectively.

In addition to their attributes, individuals maintain ties to a number of other individuals, where tie strength varies continuously in the interval 0 to 1: 0 represents a vanishingly weak tie and +1 represents the closest friendship. Unlike in other Hopfield models, we did not allow ties of repulsion, *i.e.* of negative strength, produced in situations when the net agreement in attributes between agents was negative.

During the simulation, the states of the variable attributes and zeal are affected by peer pressure. We closely follow other simulations that use Hopfield-network and assume that the agents find the average state of their friends, and move closer to that average. The probability of moving to that state depends on the strength of their ties to their peers as well as their individual pressurability constant.

With each time step in the simulation (“tic”), ties grow stronger or weaker, start or break as determined by several social processes, most importantly, through homophily (McPherson, Smith-Lovin, and Cook 2001). Namely, ties between individuals grow stronger when they have many common attributes. As an example, suppose there are 6 attributes in i and j (including zeal), and the states of i and j are (1,1,-1,1,1,1) and (1,-1,-1,1,1,1), respectively, where the last 1 indicates that they are both radicals. Furthermore, suppose none of the attributes have elevated salience. Since they agree on 5 attributes and disagree on 1, the ultimate tie strength between them would be (5-1)/6 = 0.66. If the absolute strength of a tie falls below 0.01, it is disconnected. In most simulation we allowed that ties can grow stronger over time, to represent increased familiarity and trust between the agents. However, we did not find this process had an interesting effect on radicalization, acting like a break on formation of new ties.

Whereas the above dynamics is closely modeled on previous work in Hopfield networks, magnets, transitive tie formation and attrition represent extensions of the framework[13]. Upon joining the simulated society, each agent explores the magnets and decides which ones to attend. Operationally, individuals are randomly assigned to a number of magnets, with each magnet receiving a fraction of the targeted population equal to the *magnetic exposure* parameter. Then, each tic the individual has a certain probability of visiting the magnets he/she is enrolled in and thus has a certain probability of meeting one random individual also attending that magnet. A friendship would be formed only if they are compatible (increase homophily) according to the "never swap a good friend" rule: the net agreement must be positive (4=5-1 in the above example) and either the individual is below his/her degree budget, or the new friend offers better agreement than the least agreeable existing friend. In the latter case, the old friend is dropped. Friendships may facilitate the formation of additional ties with other individuals through *transitivity*. Like magnets, transitivity allows ties to form *de novo*: each individual X, who has ties to Y1 and Y2 would act like a magnet and create a tie between Y1 and Y2. Such a transitive process was often observed in radical cells (Sageman 2004). We allowed a tie to form only at the expense of existing ties of Y1 and Y2 according to the "never swap a good friend" rule.

Our simulation also examines how demographic attrition affects radicalization. For simplicity, we assume that if an individual leaves the simulation, their friends would lose all ties to them. Attrition is modeled by randomly removing a certain fraction of the population at each tic, and replacing them with new individuals that on average have the same demographic characteristics as the rest of the population. Thus, population size is maintained constant.

[13] Indeed, rather than assuming that all agents are connected to each other (Macy et al. 2003), we explore the connection-forming process in detail. In our simulation, new connections form as a result of either magnets or introductions by friends.

In order to identify the most sensitive social parameters for each radicalization and deradicalization scenario, we generated 6000 different simulated societies. Each society was created based on parameter values that were sampled from parameter distributions which we estimated. The samples were then created using the efficient Latin Hypercube Sampling technique (McKay, Beckman, and Conover 1979; McKay 1992; Kleijnen 2008). In each society the radical population was different, in some cases meeting the definition a radicalization or de-radicalization scenario.

To measure the effect of any parameter *q* on radicalization we compared two distributions: distribution D' for the values of *q* in simulations that achieved a particular scenario (e.g. *Lone Wolfs*), and distribution D for the rest of the simulations. The difference in the means of *q* was standardized, and the distributions compared using the Mann-Whitney test. A large and significant difference implies that *q* contributes to a given scenario[14].

In order to facilitate critical review and further development, we have made the software and simulation data available online[15]. The source code is freely available for download and modification, requiring only the free open-source Repast Toolkit (North, Collier, and Vos 2006).

ANALYSIS AND FINDINGS

We now check the validity of the model based on a unique dataset of home-grown self-starter terrorism. We then analyze the effect of various model parameters on radicalization scenarios.

[14] In addition to the distribution analysis we examined sensitivity using other techniques. In particular, we fitted rank order regression models and quantile regression models. The analysis was consistent; the most sensitive parameters were consistently identified.

[15] The Internet link has been redacted to conform to JCR's policy of anonymizing the manuscript. The software can be transferred electronically to the reviewers (via the editor) if s/he requests it.

Validation the Model and Sizes of Empirical HST Cells

A simulation model carries value to the extent to which its predictions correspond to empirical reality. However, such correspondence cannot be carried out completely because empirical data is not available for every model prediction e.g. levels of clustering or isolation for radicals are very hard to assess. Fortunately, it is possible to assess the predicted cell sizes and so to partly validate the model. Therefore, we collected data on cases of HST in the United States and Western Europe from January 1990 through March 2010. Databases such as Lexis-Nexis as well as various lists[16] and datasets[17] were combed for cases of self-starter terrorism. Coders were instructed identify a terrorist cell as self-starter if the following conditions are met: 1) its members are *not* part (or have *not* been part of) a professional terrorist organization (PTO) AND 2) have *not* received assistance (e.g. financial, training, or logistical) from a PTO; 3) do not receive commands from the PTO. A PTO is defined as a terrorist organization that existed and was active for more than three years, had more than ten members, and engaged in more than three attacks that resulted in casualties or physical damage. The final dataset that we compiled contained 110 cells, ranging from 1 to 13 individuals (237 individuals total) and involves three types of movements: Islamism, Extreme Environmentalism, and various Right-Wing groups (e.g. anti-government, hate-crime, anti-abortion)[18].

These movements are the key practitioners of leaderless resistances contrasted to the more traditional organizational/pyramidal terrorism. Leaderless terrorism is not guided by a central leader or even a coterie of leaders. Rather attacks are organized from the bottom up.

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[16] Southern Poverty Law Center, FBI Chronology of Domestic Terrorism, Anti-Defamation League as well as other sources were used to compile a list of cases.

[17] The datasets we used were compiled by Alan Krueger (Krueger 2008), Steven Brooke and Robert Leiken (Leiken and Brooke 2006) , American Terrorism Study (ATS 2010) and the Global Terrorism Database (Anonymous2010).

[18] There were a few rare cases of entrapment or plea deals where judgment calls had to be made regarding inclusion. Space limitations in the paper do not allow us to discuss all the criteria used. Interested readers should contact the authors for details.

Such movements contain a large number of cultural entrepreneurs who do not carry attacks themselves, but instead promulgate cultural repertoires to sympathetic individuals who are willing to act. Cultural entrepreneurs produce the literature, tactics, and ideas and "put these ideas out there". They count on sympathizers who they do not personally know to commit violence using these cultural tools. This division of labor is deliberate and is critical to sustaining home-grown self-starter terrorism.

Figure 2 about here

The data we collected on those groups is summarized in Figure 2. The figure suggests that Islamic and Right-Wing cells tend to be smaller than the Environmental cells (mean cell sizes 2.0, 1.4 vs. 3.2). The distributions of Right-Wing and Environmentalist cells are significantly different on the Kolmogorov-Smirnov test ($p<0.01$) but the tests show no significant difference in other pairs ($p>0.05$).

This data was then compared to the prediction generated by the simulations. We wanted to see if the average size of the simulated cells matched the size of the cells observed in practice. In the simulations we define radicals quite broadly, as anybody with a high zeal level, regardless of their involvement in violent activities. Clearly only a fraction of the radicals attracts police attention or become violent. This comparison is possible only to the extent that the shift to actual violence does not depend on the size of the grouping in a significant way.

Figure 3 shows the empirical and predicted distribution of cell sizes. Visually the simulated data shows fewer 1-person cells and more 2-person cells compared to the empirical values, but otherwise they are quite similar. Indeed, the Kolmogorov-Smirnov two-sample test

has a p=0.49, suggesting that the distributions are not significantly different. To compute the difference between the distributions, we computed the Hellinger distance score, H. In general, H ranges from 0 to 1; in this case $H = 0.05$, suggesting that the distributions are indeed close. We made sure the result was not an artifact of our estimate for the distribution of average degree: the correlation coefficient of average degree with the mean cell size was only 14%.

Figure 3 about here

The correspondence of the empirical and predicted distributions shows that the model may be tapping into reality to some extent. Furthermore the model offers explanations of specific patterns in the sizes of cells between the three types of self-starter groups in our dataset. Consider Figure 3 - environmentalist groups tend to form larger cells akin to Wolf Packs, while Islamist and rightwing groups tend to form Lone Wolfs. One reason for this could be that Islamist groups are often composed of immigrants or their children who don't have dense social networks. Right-wing groups who are anti-government are more anti-society generally and more suspicious of collectivities. They are less likely to frequent magnets or to have dense social networks. They are independent individualists and hence would probably eschew having many friends.

Members of environmentalist movements display the opposite trend, that of being embedded in the broader community and connected to its institutions. Part of the reason is that those groups tend to focus on property damage rather than on harming or killing, leading to greater societal acceptance of such actions as well as greater support (in our model, greater fraction of initially radicals). Extreme environmentalist groups can also form radical magnets

without being severely sanctioned.

Cell size is also determined by certain tactical requirements. Environmentalists tend to focus on non-violent activities, which require more people to be effective. In contrast, Lone Wolf right-wingers tend to have easier access to weapons and explosives, especially in the US: they don't need large cells to organize and train for the operations. One terrorist wielding a gun can cause many fatalities, while many more people are needed to coordinate and execute a sabotage campaign.

Radicalization Scenarios and Trajectories

The findings reveal the systematic importance of certain parameters in the production of the different scenarios. While some of the results are consistent with expectation, some are counterintuitive (both positive and negative findings). Greater attention will be paid to the latter type of results. Only significant parameters as well as effect sizes close to or above 0.2 will be considered. The complete summary of the simulation results are shown in Figures 2 and 3. While Figure 4 shows the effects of the parameters on individual metrics of radicalization, Figure 5 shows the effect on radicalization and de-radicalization scenarios.

Figure 4 about here

Figure 5 about here

Lone Wolfs

The key parameters associated with creating Lone Wolfs are fraction of (initially) radicals, average degree, and transitivity. The fraction of initially radicals (FIR) refers to number of radicals divided by the size of the population, excluding people that were radicalized through

peer pressure. FIR is negatively related to Lone Wolfs because in a larger radical subpopulation, Lone Wolfs are more likely to coalesce into Wolf Packs. Average degree ("Degree" in Figs. 2 and 3) has a strong negative relationship to Lone Wolfs. The explanation is quite intuitive as having less connections means that one is more likely to be alone. A similar logic applies to transitivity**.** Surprisingly magnets did not contribute to the scenario. One might have imagined that radical magnets would create bonds between isolated radicals (a negative effect on Lone Wolfs). However, magnets are not able to surmount the basic sociological conditions that lead to Lone Wolfs: a small radical population and a low number of ties (degree).

Wolf Packs

The key parameters associated with creating wolf packs are radical magnets, fraction exposed to magnets, fraction of initially radicals, transitivity, and pressurability. The role of pressurability is most interesting. Recall that pressurability controls how easily can a node be "converted" when surrounded by opposite-minded friends. The positive effect of this parameter is counterintuitive because one might think that it should be the radicals who would be overwhelmed by the much larger non-radical population. Instead the following process likely dominates. Radicals form pockets where they are the majority and also bring some moderates into their fold.

Increasing the number of neutral magnets, the average degree, or the attrition rate tends to reduce Wolf Packs. Neutral magnets have a negative effect because they introduce radicals to non-radicals, who generally outnumber radicals. The non-radicals occupy the slots on the radical's degree budget thereby preventing triadic closure (see Fig. 6). The negative effect of average degree is perhaps surprising, since radical cells themselves require connections to many

radicals. While true, having many friendships hurts Wolf Packs in two ways (Fig. 4). First, the more friends a radical has the more likely one of those friends would be a moderate, reducing Isolation; Second, it becomes more difficult to achieve high Clustering.

Attrition (population turnover) facilitates the destruction of all social ties, and has the beneficial effect of damaging radical packs and reducing radicalization. Radicals who arrive at the community tend to form ties with non-radicals, at least before they travel the long road to finding like-minded persons.

Trapped Wolfs

Broadly speaking, Trapped Wolfs are created by processes that destroy ties in Wolf Packs or that create ties in Lone Wolfs. Magnets play an important role in the trapping of Lone Wolfs: Trapped Lone Wolfs increase with an increase in neutral magnets, a decrease in radical magnets, and a decrease in the fraction exposed to magnets. Structural parameters too play a large role, perhaps affecting this scenario more than the other four scenarios. Increasing the Fraction of Initially Radicals tends to decrease the number of Trapped Wolfs. When more radicals are found, they make larger cells, ultimately creating Wolf Packs and Trapped Wolf Packs.

Network measures such as average degree exert a positive effect while transitivity exerts a negative effect. Furthermore we see that only one individual factor matters, diversity, having a negative effect. Once again we see the deradicalizing effect of high average degree and neutral magnets as both parameters facilitate a process of radical encapsulation.

Trapped Wolf Packs

To understand how Trapped Wolf Packs are created, recall that they have high cell size,

high clustering, and low isolation. Parameters operative on those metrics through two broad paths to bring about this de-radicalization scenario: the parameters may encourage *encapsulation* of existing radicals or they may prevent their formation in the first place - *prevention*. For example neutral magnets decrease the formation of the packs of radicals in the first place by decreasing clustering of radicals, instead introducing them to non-radicals.

Much like the magnets, attrition exerts a negative role through prevention rather than encapsulation. It decreases clustering by breaking up radical cells hence converting Trapped Wolf packs to trapped Lone Wolfs. Average degree and transitivity on the other hand operate via encapsulation: they encourage the surrounding of radical cells by non-radicals. Pressurability exerts a negative effect on trapping radical cells because it allows for the easy conversion of moderates who would otherwise act as encapsulators.

Summary of Simulation Findings: Defanging the Wolves

We can also see broader patterns of how the various model parameters or social conditions radicalize and de-radicalize. For example, neutral magnets deradicalize by facilitating mixing with non-radicals and then encapsulation. Attrition plays a similar role, disrupting radical cells from forming

However, the most powerful parameter is average degree (i.e. network density). Increasing network density traps wolfs and prevents Wolf Packs from forming. Human beings have a limited set of friends that they can maintain strong ties with and if those slots are filled up, certain other friendships may not form and clustering may be prevented. In the context of radicalization this manifests itself in a dynamic we call *occlusion*. Occlusion occurs when

moderates[19] fill up the degree budget of a potential radical and prevent triad closure from obtaining, as shown in Figure 6. Let's assume that each node has a degree budget of two. Ordinarily, C would form a tie with B via the principle of transitivity, but because C has a tie with U, he does not form the tie with B. (Most people have a budget higher than two and so this is meant as the simplest illustration of the process.)

Figure 6 about here

DISCUSSION: TOWARDS AN ECOLOGY OF RADICALIZATION

Our simulation study proposes an ecological perspective on radicalization that considers how the social ecology leads to the self-organization of radical networks. Our approach complements studies that examined the factors that radicalized particular individuals.

The findings from this approach are suggestive for new directions in understanding radicalization and have some important policy implications. Perhaps the most important finding is that radical social sites are highly important in contributing to radicalization. At the same time, neutral sites like paintball clubs and sports leagues contribute to *de-radicalization* even though they were thought of as central in explanations of how the terrorist groups formed (e.g. House of Commons Report, 17)[23]. Indeed, one may suppose that any process that promotes homophilous ties would lead to greater radicalization. However, such a supposition ignores the range of

[19] The process can also theoretically occur with radicals. A radical from a different cluster may occupy the degree budget of another radical preventing triadic closure.

[23] Also see this popular press article that claims that gyms rather than mosques are the breeding ground for terrorism(O'Neill 2006).

possible attributes that can form the basis of ties. Thus, while neutral magnets do form some ties between radicals, their net effect is to de-radicalize because they encapsulate radicals with non-radical ties. Furthermore, pacifist magnets such as "reconciliation societies" do not appear to be effective against radicalization since they draw from people who are already committed to their ideology, that is they preach to the choir. The most effective positive magnets may have to be those that are disguised as radical magnets so that they can draw in unsuspecting radicals and gradually socialize them. For example, an organization that offers community paintball – a high action type of activity (that may draw in individuals with radical beliefs), which in subtle ways introduces lessons of coexistence by pairing Islamists and non-Muslims on the same team.

Aside from magnets, network density is one of the most critical factors in de/radicalization. If a society wants to prevent radicalization it is not necessary to get the radicals to disavow their beliefs; a more practical objective is to integrate those with radical views bymaking their networks dense with non-radicals. Indeed this strategy seems more effective than arresting the radicals i.e. attrition.

Pressurability (i.e. impressionability through peer pressure) matters for radicalization. One might have expected that having many pliable individuals would lead to de-radicalization since radicals tend to form a small minority that should be easily overwhelmed by societal norms. Instead, radicals form closely-knit subcultures where they constitute the majority and are able to convert non-radicals that are closely-tied to them.

The results of the model suggest some policies and locations where one may want to focus resources. For example, we found that radicalization is more likely to occur in cities with low migration rates and low number of social venues. At the very least, it points to possible targets of police resources. Under certain conditions (e.g. low fraction of radicals, high average

degree, and low transitivity) Wolf Packs are unlikely to form and the main threat comes from Lone Wolfs. Therefore, when those conditions apply, law enforcement is better focused on individuals that operate by themselves rather than hunting for large groups.

One natural question often raised in the context of our work is the significance of the Internet for radicalization. Do websites, particularly those that promote radical ideologies, create forums for radicals to exchange information, and make acquaintance with one another; are they a type of "cyber-magnet"? In our opinion Internet sites can serve as magnets only if they facilitate high-trust ties. Arguably, few real friendships are made through such a virtual medium because online persons might be mistrusted and for good reasons: the online persons might be intelligence officers pretending to be radicals. Radicals would probably want to meet face-to-face to form a tie, certainly if they are to trust one another, and especially if they are hoping to eventually carry out an illegal activity such as an attack. Indeed, we saw few empirical cases where cells were formed online. Hence, radical websites may provide general information to their readers, but to actually rewire ties, it would be necessary for the website to be localized in a specific community. Therefore, website's main role is to nourish the radical movement rather than to form individual cells.

Finally, one important idea is the notion of radicalization trajectories. Almost every terrorist group was at one point just a "bunch of guys". What conditions influence whether a cell sprouts into a full-fledged organization or whether it withers away after a few attacks? While our model can't make point predictions, it can shed some light on what the ingredients of the influential conditions look like. The model strongly suggests that a long-lasting organization must initially be like a Wolf Pack scenario: many large, isolated, and clustered cells. We saw that such cells are the result of a high fraction of radicals, low degree budgets, high transitivity, a

large number of radical magnets, and high pressurability. Therefore factors that contribute to the increasing of these parameters nurture the formation of terrorist organizations.

FUTURE DIRECTIONS

While we have considered a large set of effects, the model is capable of expanding into a number of other directions. In particular the model can benefit from adding additional parameters and processes, as well as additional empirical validations some of which we identify below.

Based on our findings it should be possible to construct a risk index for metropolitan areas across Western Europe and the United States. This index will give for each area the likelihood of producing home-grown self-starter terrorist cells, and will be based on the values of the parameters in that area. For this index to be estimated, it would be necessary to obtain a lot of the empirical data, but we believe that the task is feasible because it would be mainly based on existing demographic and economic tables.

The model could be extended to include other important parameters and processes. One interesting extension is to consider the effect of charismatic individuals (e.g. such as clerics), both radical and non-radical. Unlike magnets who simply introduce individuals, those individuals would increase/decrease the radicalization of individuals who chance upon them. The effect is not predictable in advance because individuals have different susceptibility to such authority figures and may even be repelled by them.

Another important process is the formation of weak ties. Our model allows ties to carry different weights, thus affecting the influence it carriers on its participants. However, some ties may never become psychologically demanding while playing an important social role. Such "weak ties" likely play an important role facilitating terrorism, particularly connecting radicals

with expertise and specialized equipment. That said, terrorism requires full commitment to what is a minority cause and a break with normal society. Because of that, self-starter radical cells must be based on the back-bone of strong ties between their members.

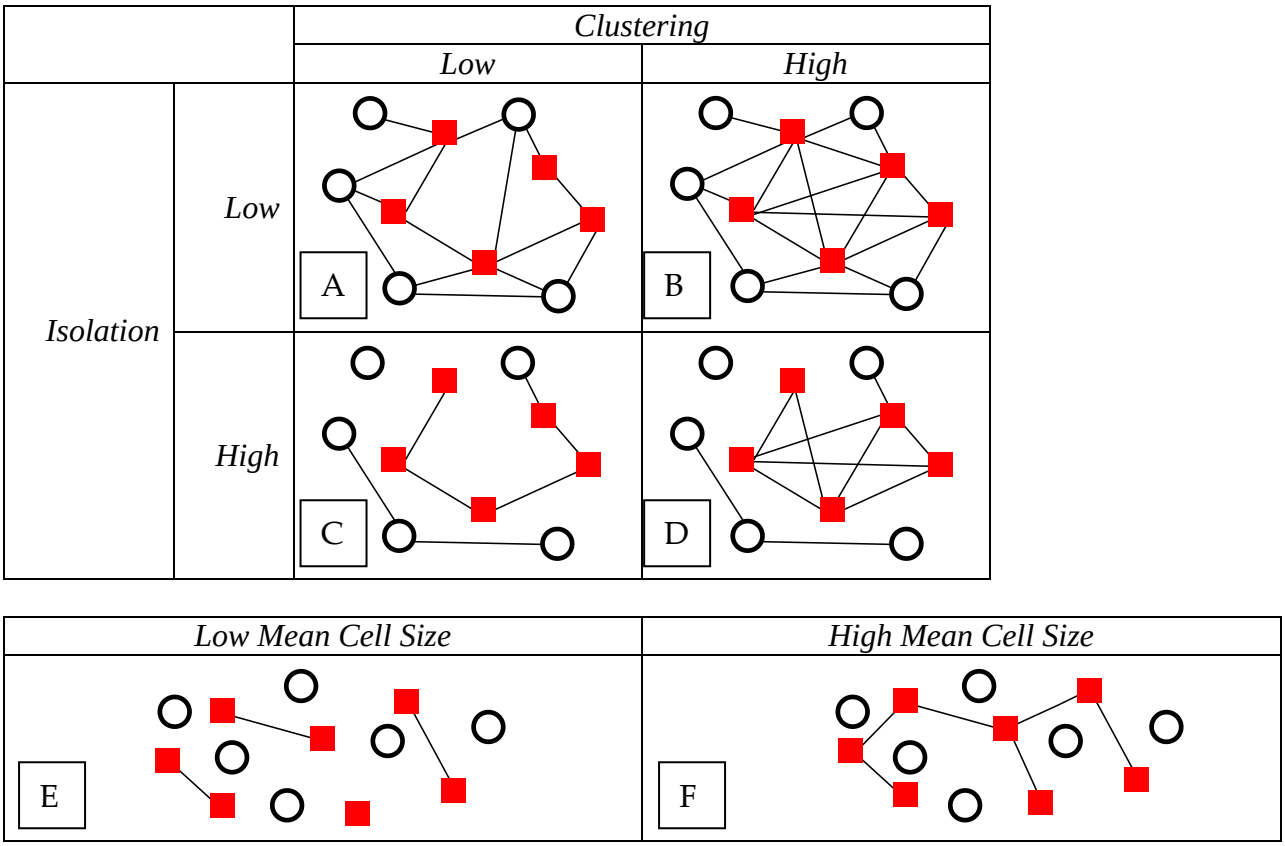
In principle the processes we considered are much broader than self-starter terrorism or even political violence, and take place whenever any set of individuals who hold extreme views seek one another's company. For example with appropriate adjustments the model and measures of radicalization can be extended to hate groups, cults, marginal social movements, or even gangs.

CONCLUSION

The paper proposed and analyzed a relatively simple, empirically-based dynamic model that treated radicalization as a network ecological process. It showed why, when and how certain factors such as low attrition, low average network degree, and the presence of certain kinds of meeting sites can facilitate self-radicalization, while other plausible factors such as community size have little effect. We hope to have made three key contributions. First, we operationalized radicalization as a network process and distinguished between several kinds of radicalization scenarios. Second, we introduced an agent-based model and characterized the structural, individual, and network factors that contribute to radicalization as well as deradicalization. Furthermore we validated some of the model's predictions empirically. Third; we extended radicalization theory by identifying specific network mechanisms such as occlusion, encapsulation, and magnet-rewiring by which radicals self-organize. We hope to have made

contributions to the methodological approach as much as to the theoretical substance of the topic. Our findings paint a complex picture of an important process, raise new and interesting questions, and suggest future research directions.

FIGURES AND TABLES



Note: Red squares represent radicals and black circles represent non-radicals.

Figure 1: Three network-based measures of radicalization: Isolation, Clustering and Mean Cell Size.

Table 1

Correlation between the three Metrics of Radicalization over the simulation data. Standard errors are shown in parentheses. The metrics are largely independent.

	Isolation	Clustering	Mean Cell Size
Isolation	1		
Clustering	0.0416 (0.0013)	1	
Mean Cell Size	0.0264 (0.0408)	-0.0617 (0.0000)	1

Table 2

Definitions of social scenarios. The percentages refer to percentiles of the simulation.

Scenario	Isolation	Clustering	Mean Cell Size
Lone Wolfs	Top 50%	Bottom 50%	Bottom 50%
Wolf Packs	Top 50%	Top 50%	Top 50%
Trapped Wolfs	Bottom 50%	Bottom 50%	Bottom 50%
Trapped Wolf Packs	Bottom 50%	Bottom 50%	Top 50%

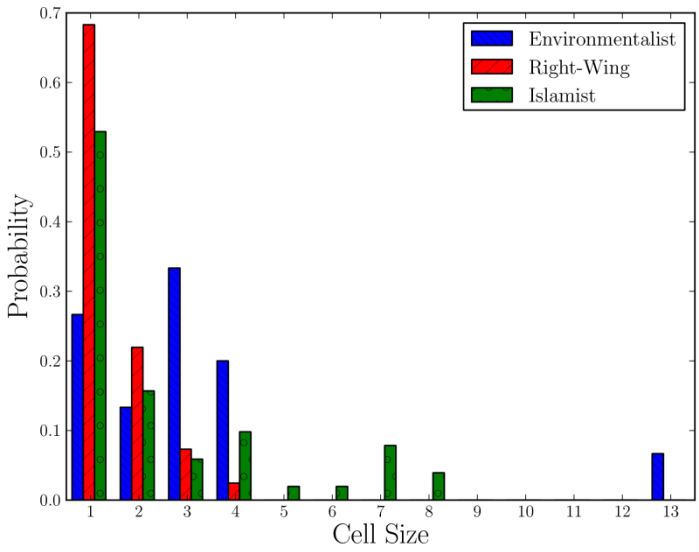


Figure 2:Empirical distribution of cell sizes. Right-Wing cells tend to be small compared to Islamist and especially Envonmentalist groupings.

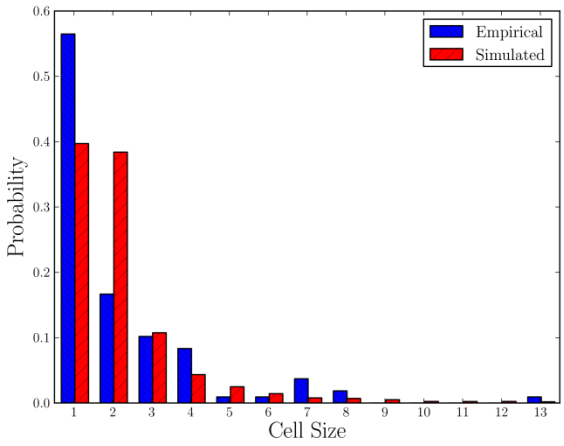


Figure 3:Distribution of cell sizes in simulation and empirical cases. The values are statistically indistinguishable, giving important validation to the model. Bin 13 contains all cells of size 13 or more.

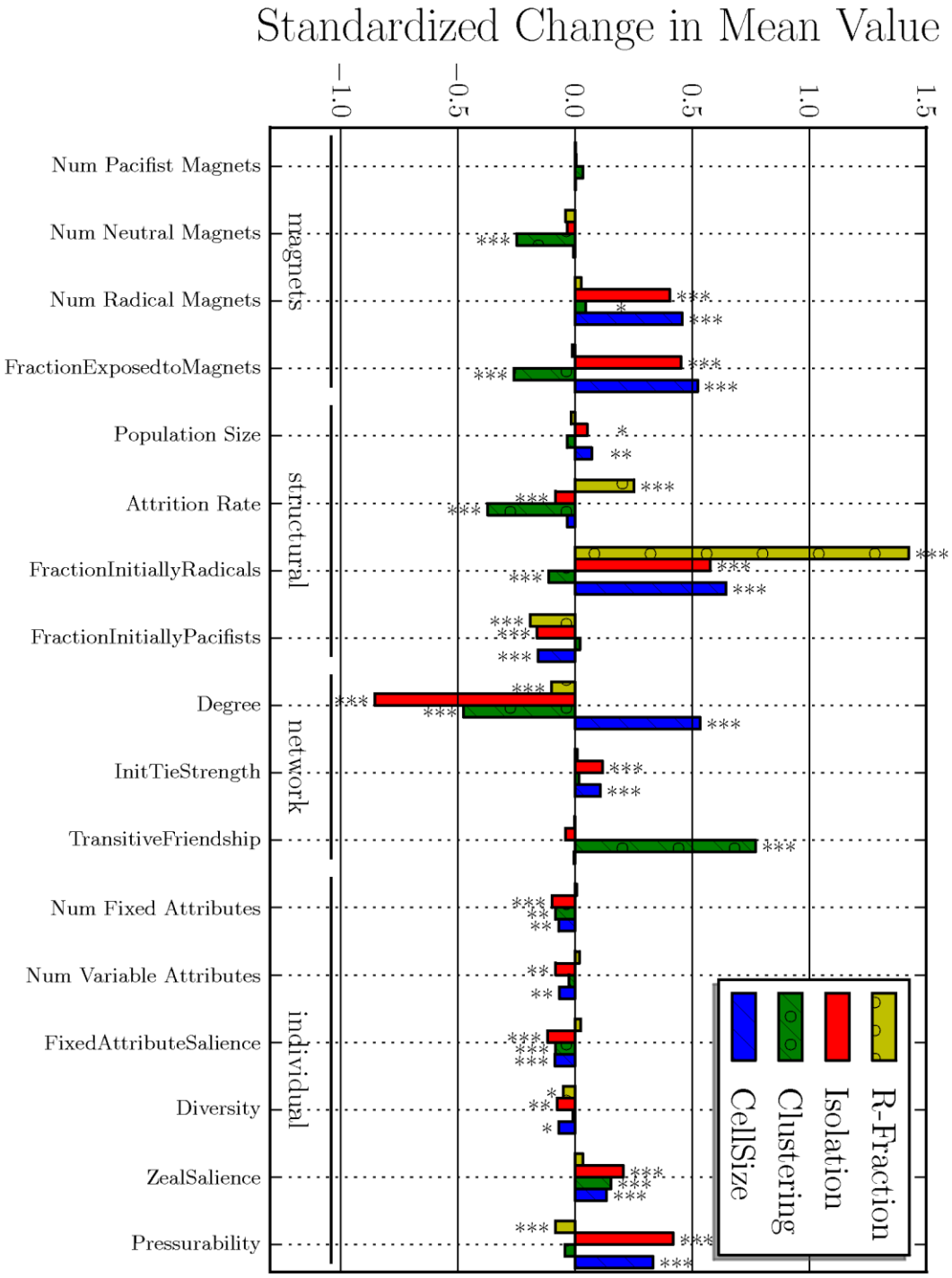


Figure 4: Direct effects of the social parameters on measures of radicalization. The four series represent simulations where each of the four measures was in the top 50% of simulations. For each parameter, a bar indicates the difference in the mean value of the parameter (mean over top 50% - mean in the full set) standardized by the standard deviation in the full set. Significance

is based on the Mann-Whitney test (*<0.05, **<0.01 and ***<0.001).

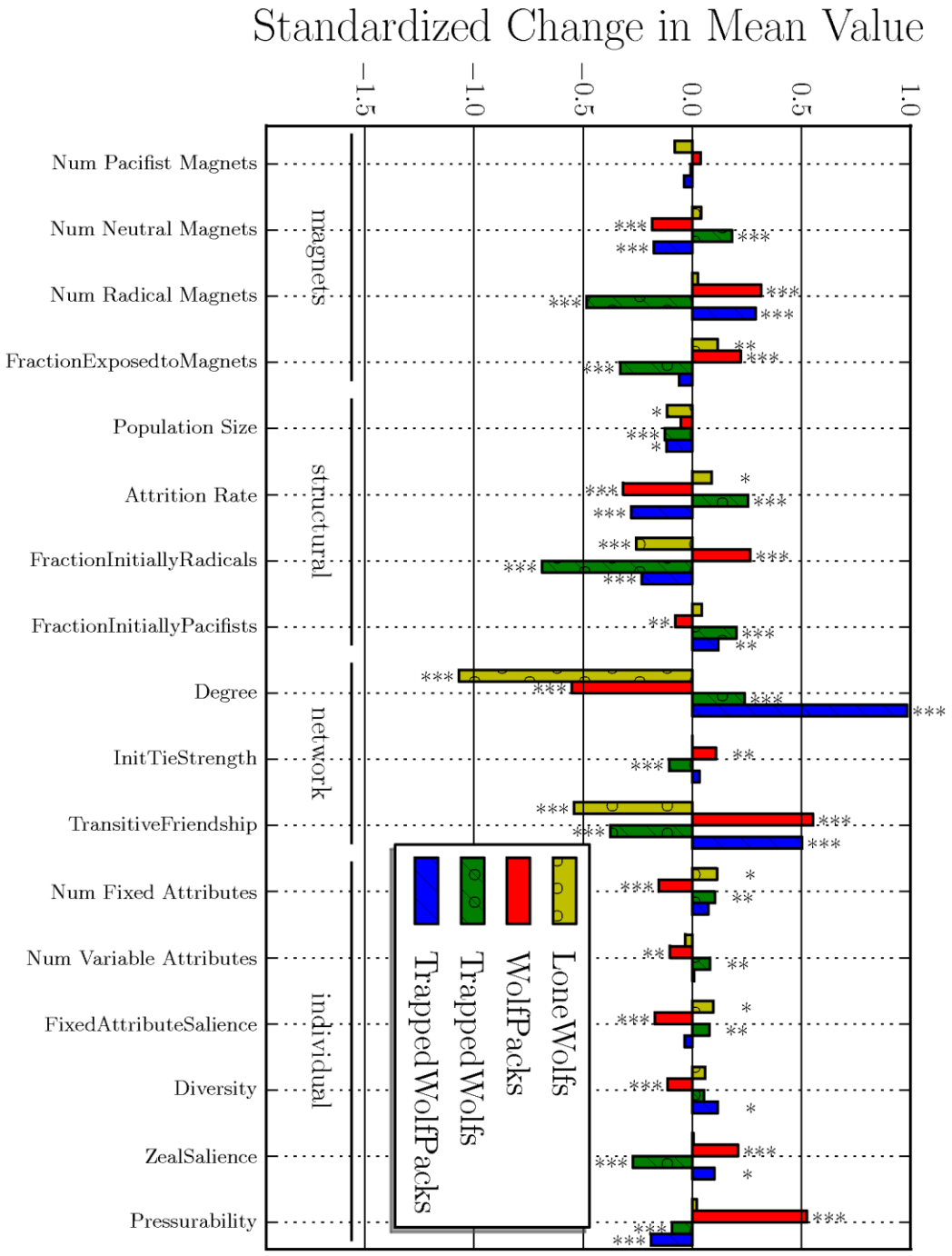


Figure 5: Scenario Analysis of the Simulation. For each parameter, the bar indicates its importance to the scenario. Namely, the bar's length is the difference in the mean value of a parameter under a scenario compared to the rest of the simulations. Normalization is by the standard deviation in the full set.

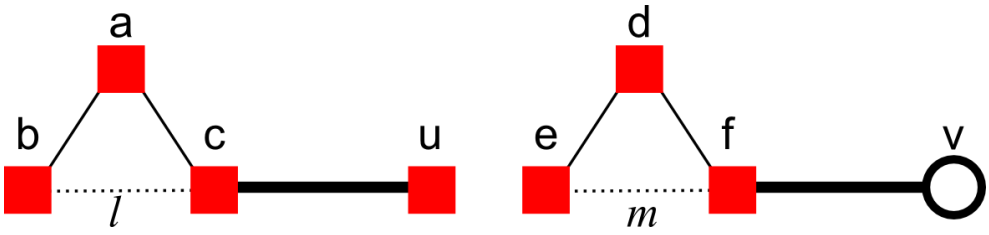


Figure 6: The Occlusion Process. The radicals a,b,c fail to form a clustered cell (edge l), as may be expected from the triad closure process, because of a strong tie with a nodes outside the cell (node u). Node c has a tie to a radical (node u), as may have been formed through an encounter in a radical magnet. A similar process disrupts a triad of d,e and f where a tie to non-radical v.

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